

Tailwind CSS exercises

Instructions

Complete the exercises using plain HTML and the Tailwind Play CDN.

For exercises that require a complete page, create the file using the name shown and open it in a browser. An internet connection is required for the Tailwind Play CDN.

Do not use:

- Custom CSS
- JavaScript
- npm or a Tailwind configuration file
- Utility classes that have not been introduced in the exercises or lessons

Different class choices are acceptable when they satisfy the stated requirements.

Section A: Setup and utility classes

Exercise 1: Find why Tailwind is not working

Create this file:

```
exercise-01/  
└─ index.html
```

The page contains valid Tailwind classes, but the browser shows unstyled HTML:

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  
  <meta  
    name="viewport"  
    content="width=device-width, initial-scale=1.0"  
  >  
  
  <title>Community notice</title>  
</head>  
<body class="bg-stone-100 p-6">  
  <main class="rounded-lg bg-white p-6">  
    <h1 class="text-3xl font-bold">  
      Community notice  
    </h1>
```

```
<p class="mt-3 text-stone-600">
  The reading room closes at 6 PM.
</p>
</main>
</body>
</html>
```

Complete the following work:

1. Identify why the Tailwind classes are being ignored.
2. Add the required Tailwind Play CDN script in the correct part of the document.
3. Reload the page and confirm that the background, spacing, and text styles appear.
4. Explain why this setup requires an internet connection.

Exercise 2: Read a Tailwind class list

Read this element:

```
<aside
  class="
    mt-6 rounded-lg border
    border-amber-300 bg-amber-50 p-4
  "
>
  Registration closes on Friday.
</aside>
```

Describe what each utility changes:

```
mt-6
rounded-lg
border
border-amber-300
bg-amber-50
p-4
```

Then answer these questions:

1. Which utility adds space inside the notice?
2. Which utility adds space outside the notice?
3. Which utility creates the border width?
4. Which utility changes only the border colour?
5. Why would `border-amber-300` not create a visible border by itself?

Exercise 3: Put each utility on the correct element

The following classes need to be added to the heading or its container:

```
rounded-lg  
border  
bg-white  
p-6  
text-3xl  
font-bold  
text-stone-900
```

Start with this HTML:

```
<section>  
  <h2>  
    Evening photography walk  
  </h2>  
  
  <p>  
    Meet at the east entrance at 5:30 PM.  
  </p>  
</section>
```

Apply the classes so that:

- The section has a white background.
- The section has a border and rounded corners.
- The section has padding on all sides.
- The heading is large, bold, and dark.
- Container styling is not placed on the heading.

After completing the HTML, explain why `p-6` belongs on the section rather than the heading.

Section B: Text, spacing, backgrounds, and borders

Exercise 4: Translate CSS decisions into Tailwind utilities

A notice currently uses conventional CSS:

```
<aside class="admission-notice">  
  <h2>Admission</h2>  
  
  <p>
```

```
    Entry is free. Food and products  
    are sold separately.  
  </p>  
</aside>
```

```
.admission-notice {  
  margin-bottom: 1.5rem;  
  padding: 1rem;  
  background-color: #fef3c7;  
  border-left: 4px solid #f59e0b;  
  border-radius: 0.5rem;  
}
```

Replace the custom class with Tailwind utilities.

Use these utilities where appropriate:

```
mb-6  
p-4  
bg-amber-100  
border-l-4  
border-amber-500  
rounded-lg
```

Do not change the text or add a `<style>` element.

Then identify the Tailwind utility that replaces each CSS declaration.

Exercise 5: Style a workshop card

Start with this HTML:

```
<article>  
  <p>Beginner workshop</p>  
  
  <h2>Build your first web page</h2>  
  
  <p>  
    Practise HTML structure and basic styling.  
  </p>  
  
  <p>Saturday, 10 AM</p>  
</article>
```

Style the card so that:

- The card has a white background.
- It has a visible stone-coloured border.
- Its corners are rounded.
- It has padding on all sides.
- It has space below it.
- The label is small, bold, and indigo.
- The heading is larger and bold.
- The description has space above it and uses a softer text colour.
- The date has space above it and is semibold.

You may use:

```
bg-white
border
border-stone-300
rounded-lg
p-6
mb-6
text-sm
font-bold
text-indigo-700
text-2xl
mt-3
text-stone-600
font-semibold
```

Place the classes on the elements whose appearance they control.

Exercise 6: Repair a broken notice

The border and padding work, but the background colour does not appear:

```
<aside
  class="
    rounded-lg border-l-4
    border-indigo-500 bg-indigo-50 p-4
  "
>
  Bring a notebook for the activity.
</aside>
```

Answer these questions:

1. Which class is incorrect?
2. What should replace it?

3. Why do the other styles continue to work?
4. What debugging approach would help you find this problem without replacing the complete class list?

Then correct the HTML.

Section C: Flexbox and Grid

Exercise 7: Arrange a title and status on one row

Start with this card header:

```
<div>
  <h2>Visual Studio Code</h2>
  <span>Installed</span>
</div>
```

Add Tailwind classes so that:

- The title and status appear on the same row.
- They are vertically centred.
- The title stays on the left.
- The status moves to the right.

Use:

```
flex
items-center
justify-between
```

Answer these questions after completing the code:

1. Which element must receive the Flexbox classes?
2. Which elements become Flexbox items?
3. Why would adding `flex` to the `<h2>` not arrange the `<span>` beside it?

Exercise 8: Change the direction of a navigation group

Start with this navigation:

```
<nav>
  <a href="#">Overview</a>
  <a href="#">Schedule</a>
```

```
<a href="#">Contact</a>
</nav>
```

Apply Tailwind classes so that:

- The links are arranged using Flexbox.
- The links appear one below another.
- There is space between the links.

Use:

```
flex
flex-col
gap-3
```

Then change the layout so that the links appear in one row.

Answer these questions:

1. What does `flex` enable?
2. What does `flex-col` change?
3. Why does `flex-col` not create a Flexbox layout by itself?

Exercise 9: Build a card Grid

Start with this HTML:

```
<section>
  <article>HTML foundations</article>
  <article>Responsive CSS</article>
  <article>Tailwind utilities</article>
  <article>JavaScript basics</article>
</section>
```

Add classes to the `<section>` so that:

- Its direct children are arranged using Grid.
- The layout contains two equal columns.
- There is consistent space between the cards.

Use:

```
grid
grid-cols-2
gap-4
```

Then answer:

1. How are the four cards placed across the rows?
2. What would happen if a fifth card were added?
3. Which element controls the Grid layout?
4. Would adding `grid-cols-2` to each `<article>` create the required layout? Explain your answer.

Exercise 10: Use Grid for cards and Flexbox inside each card

Start with this HTML:

```
<section>
  <article>
    <div>
      <h2>Quiet room</h2>
      <span>Available</span>
    </div>

    <p>Seats: 8</p>
  </article>

  <article>
    <div>
      <h2>Discussion room</h2>
      <span>Occupied</span>
    </div>

    <p>Seats: 12</p>
  </article>
</section>
```

Add Tailwind classes so that:

- The `<section>` arranges the room cards in two columns.
- There is a gap between the cards.
- Each `<article>` has a border, rounded corners, and padding.
- The title and status inside each card appear on one row.
- The title and status are vertically centred and placed at opposite ends.

Use Grid for the card collection and Flexbox inside each card.

After completing the layout, identify:

1. The Grid parent
2. The Grid items
3. The Flexbox parents
4. The Flexbox items

Section D: Responsive layouts

Exercise 11: Predict responsive behaviour

Read this class list:

```
<section
  class="
    grid grid-cols-1 gap-4
    md:grid-cols-2
  "
>
  ...
</section>
```

Answer these questions:

1. How many columns are used before the `md` breakpoint?
2. How many columns are used at the `md` breakpoint and wider?
3. Does `gap-4` stop applying at `md`?
4. Which class represents the mobile-first base layout?
5. Why is `grid-cols-1` not written as `sm:grid-cols-1` in this example?

Exercise 12: Correct a reversed responsive layout

The following class list produces two columns on small screens and one column on wider screens:

```
<section
  class="
    grid grid-cols-2 gap-6
    md:grid-cols-1
  "
>
  ...
</section>
```

The intended behaviour is:

- One column on narrow screens
- Two columns at `md` and wider
- The same gap at every size

Correct the class list.

Then explain:

1. Which class applies before any breakpoint condition is met?
2. Which class overrides the column count at `md`?
3. Why does the order of the requirements follow a mobile-first approach?

Exercise 13: Create a responsive header

Start with this HTML:

```
<header>
  <div>
    <h1>Community learning centre</h1>
    <p>Weekend workshops and study groups</p>
  </div>

  <a href="#">View schedule</a>
</header>
```

Add Tailwind classes so that:

- The header uses Flexbox.
- Its contents are stacked on narrow screens.
- The link appears below the title group on narrow screens.
- At `md` and wider, the title group and link appear on one row.
- The two sides are vertically centred at wider sizes.
- They move to opposite ends at wider sizes.
- There is a gap while the content is stacked.

Use:

```
flex
flex-col
gap-4
md:flex-row
md:items-center
md:justify-between
```

Explain why the unprefixed classes describe the narrow-screen layout.

Final exercise: Build a responsive learning-space directory

Create this file:

```
tailwind-final/  
└─ index.html
```

Use the prepared HTML below. Do not remove or rename any element.

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  
  <meta  
    name="viewport"  
    content="width=device-width, initial-scale=1.0"  
  >  
  
  <title>Learning-space directory</title>  
</head>  
<body>  
  <main>  
    <header>  
      <div>  
        <p>Community centre</p>  
        <h1>Learning-space directory</h1>  
        <p>  
          Find a suitable room for individual  
          study or group work.  
        </p>  
      </div>  
  
      <a href="#">View centre rules</a>  
    </header>  
  
    <aside>  
      Rooms marked Available can be used  
      without an advance booking.  
    </aside>  
  
    <section>  
      <article>  
        <div>  
          <h2>Quiet room</h2>  
          <span>Available</span>  
        </div>  
  
        <p>  
          A low-noise room for reading  
          and individual study.  
        </p>
```

```

        <p>Seats: 8</p>
    </article>

    <article>
        <div>
            <h2>Discussion room</h2>
            <span>Occupied</span>
        </div>

        <p>
            A shared room for group activities
            and project discussions.
        </p>

        <p>Seats: 12</p>
    </article>

    <article>
        <div>
            <h2>Computer room</h2>
            <span>Available</span>
        </div>

        <p>
            Desktop computers with internet
            access for guided practice.
        </p>

        <p>Seats: 10</p>
    </article>
</section>
</main>
</body>
</html>

```

Part 1: Connect Tailwind

Add the Tailwind Play CDN script inside `<head>` .

Part 2: Style the page

Apply these requirements:

Page

- Use a light stone background.
- Use dark stone text.
- Add padding around the complete page.
- Keep the main content readable by limiting its maximum width.
- Centre the main content horizontally.

You may use:

```
bg-stone-100
text-stone-900
p-6
max-w-5xl
mx-auto
```

Header

- Arrange the header content with Flexbox.
- Stack the title group and link on narrow screens.
- Place them on one row at `md` and wider.
- Align them vertically and move them to opposite ends at wider sizes.
- Add space below the header.
- Style the small label, main heading, description, and link so their roles are clear.

Notice

- Add a pale amber background.
- Add a thicker amber left border.
- Add padding and rounded corners.
- Add space below the room directory.

Room directory

- Use Grid.
- Show one column on narrow screens.
- Show two columns at `md` and wider.
- Use a consistent gap between cards.

Room cards

- Give each card a white background.
- Add a border, rounded corners, and padding.
- Place each room title and status on one row.
- Align them vertically and move them to opposite ends.
- Add suitable spacing between the title area, description, and seat count.

Do not use custom CSS.

Completion checks

Confirm that:

- Tailwind styles load in the browser.
- The page content does not touch the viewport edges.
- The header is stacked on a narrow screen and changes to a row at `md`.
- The directory uses one column on narrow screens.
- The directory uses two columns at `md` and wider.

- Flexbox is applied to the parents of the items being arranged.
- Grid is applied to the parent of the room cards.
- The notice is visually distinct from the page background.
- Each card has visible spacing, a border, and rounded corners.
- No custom CSS or JavaScript has been added.